

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&E) (Credit System) Degree Examinations**

January 2023

20EE501 – LINEAR INTEGRATED CIRCUITS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

2) Use of op-amp datasheet and standard resistor, standard capacitor sheet is permitted.

Unit – I**Marks BT* CO* PO***

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|-------|--|---|-----|---|---|
| 1. a) | Explain the following terms with reference to op-amps
(i) Input offset voltage (ii) Gain Bandwidth Product (iii) Slew Rate | 6 | L*1 | 1 | 1 |
| b) | A high Z_{in} capacitor coupled voltage follower with $R_L=12k\Omega$ is to be designed using a 741 op-amp. Determine suitable component values to give a 70Hz lower cut-off frequency. | 6 | L3 | 1 | 3 |
| c) | Explain the circuit instability due to stray capacitance. List some of the precautions taken to stabilize the op-amp circuit. | 8 | L1 | 2 | 1 |
| 2. a) | Explain the capacitor coupled inverting amplifier circuit. Show the design steps to be followed. How to set the lower and upper cut-off frequencies? | 7 | L2 | 1 | 2 |
| b) | Define and explain the terms CMRR. If the differential voltage gain and common mode voltage gain of a differential amplifier is 48dB and 2dB respectively, calculate CMRR. | 5 | L2 | 1 | 2 |
| c) | Discuss the effect of load capacitance on Op-Amp circuit stability. Explain the method to compensate the load capacitance. | 8 | L2 | 2 | 2 |
| 3. a) | Using a 741 Op-Amp, design a high Z_{in} non-inverting amplifier to operate with single polarity power supply of +36V. Load resistance is to be $10k\Omega$, lower cutoff frequency is 100Hz and gain is to be equal to 20. | 7 | L3 | 1 | 3 |
| b) | Explain the capacitor coupled difference amplifier and its design. | 6 | L1 | 1 | 1 |
| c) | Draw the uncompensated op-amp gain & phase response and explain. Define and explain the terms Phase margin. | 7 | L2 | 2 | 2 |

Unit – II

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|-------|--|---|----|---|---|
| 4. a) | Explain the working of dead zone circuit. Show how it can be used to design half wave precision clipper circuit. Draw the waveforms at various stages. | 7 | L2 | 3 | 2 |
| b) | Design a noninverting Schmitt trigger with following specification $UTP = +4V$, $LTP = -2.5V$, use a 741 op-amp with $V_{CC} = \pm 20V$. Consider rail-rail operation. Draw the neat circuit diagram. | 6 | L3 | 3 | 3 |
| c) | Explain the working of a two stage third order active low pass filter and explain its operation. Give the design steps and selection of stage frequencies. | 7 | L2 | 4 | 2 |
| 5. a) | A $\pm 1.5V$ signal is to be sampled by a Sample and Hold circuit. The holding time is to be $700\mu s$, and the output accuracy is to be 0.3%. LM108 with $I_{B(max)}=2nA$ and the FET with 25Ω channel resistance and $300nA$ gate-source reverse leakage current is to be used. Draw the designed the circuit and determine minimum acquisition time. | 7 | L3 | 3 | 3 |

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|-------|--|---|----|---|---|
| b) | Explain the working of inverting Schmitt Trigger circuit and draw the waveform. Show the design steps to be followed. | 6 | L1 | 3 | 1 |
| c) | With a neat block diagram explain each block of Phase Locked Loop. Mention various applications of PLL. | 7 | L2 | 4 | 2 |
| 6. a) | With a neat circuit explain an Op-Amp based precision clamping circuit. Give the design steps. | 6 | L2 | 3 | 2 |
| b) | Sketch an op-amp based astable multivibrator and explain its working. Draw the circuit input output waveforms. Give the design steps and equation of frequency of operation. | 7 | L2 | 3 | 2 |
| c) | Design a first order high pass active filter circuit to have a cutoff frequency of 5 kHz. Use LM108 op-amp and estimate the highest frequency that can be passed. Draw the circuit and frequency response. Unity gain frequency is 1MHz. | 7 | L3 | 4 | 3 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | A 555 timer based monostable multivibrator circuit uses a 15 V supply, determine suitable R_1 and C_1 values to give 1 ms output pulse each time the circuit is triggered. Draw the circuit diagram. | 7 | L2 | 5 | 2 |
| b) | Explain the working of a free ramp generator using 555 timer. | 6 | L1 | 5 | 1 |
| c) | Elaborate the working of a 555 timer based square wave oscillator. Give the design steps. | 7 | L2 | 5 | 2 |
| 8. a) | Explain the working of precision voltage regulator. Give the detailed design steps to be followed. How to calculate the performance parameters? | 10 | L2 | 5 | 2 |
| b) | A 12V, 40mA LM723 regulator is to be designed with short circuit protection. The short circuit current is to be 45mA. Calculate resistor values and IC power dissipation under normal and short circuit condition. Draw the circuit diagram. | 10 | L3 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
